MCQ A · differentiation and convexity

After S29; the function is differentiable on each interval considered.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. If $f''(x) > 0$ on an interval, f is:

A. decreasing
B. convex
C. concave
D. constant

2. The derivative of e^{1+x^2} is:

A. e^{1+x^2}
B. $2xe^{x}$
C. $2xe^{1+x^2}$
D. $1 + 2x$

3. A tangent to the graph of a convex function is:

A. always above
B. always below on the interval
C. parallel to the axis
D. does not exist

4. If $f'(x)$ changes sign from $+$ to $-$ at a, then f :

A. has a minimum at a
B. has a local maximum at a
C. is convex
D. is not continuous

5. The function $x \mapsto |x|$ is:

A. differentiable at 0
B. not continuous at 0
C. continuous but not differentiable at 0
D. concave on $\mathbb{R}$

MCQ B · continuity and the IVT

After S33.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. To apply the IVT to $f(x) = 0$ on $[a; b]$, first check:

A. that f is continuous on $[a; b]$

B. that f' is zero

C. that $a = b$

D. that f is affine

7. If $f(a) < 0 < f(b)$ and f is continuous on $[a; b]$, then:

A. f has exactly one root

B. f has at least one root

C. f is increasing

D. $f(a) = f(b)$

8. To prove that a solution is unique, it is also enough, for example, that f be:

A. bounded

B. strictly monotonic

C. even

D. differentiable at a only

9. The bisection method preserves at each step:

A. two bounds of the same sign

B. an interval containing a root with opposite signs at the endpoints

C. a horizontal tangent

D. an exact value of the root

10. A graph of f'' below the axis means that f' is:

A. increasing

B. decreasing

C. positive

D. zero

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every continuous function has a root.

2. If f is convex, then f' is increasing where it is defined.

3. An approximate value obtained by bisection is a proof of existence.

Review of prior work

I can link the sign of f'', convexity and variations of f'; isolate, then justify a solution. **To watch out for:** claiming uniqueness without monotonicity; confusing a graphical conjecture with the IVT.